

Inclusive Educational Software System: Emotionally Expressive Reading for Children with Disabilities (Blind, Deaf, Cognitive Impairments)

Project ID: 25-26J-407

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
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August 2026

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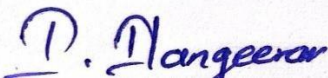
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DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



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Signature of the Supervisor

Ms. Sanjeevi Chandrasiri

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Date



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.....27/08/2025.....

Date

ABSTRACT

Children with cognitive disabilities often face significant challenges in understanding and engaging with complex educational text, which can hinder their learning experience and emotional connection to the content. Existing educational tools often lack the ability to adapt dynamically to individual comprehension levels or to convey emotions in a way that resonates with these learners. This research addresses this gap by proposing an intelligent, emotion-aware storytelling and learning support system that simplifies and summarizes educational text while detecting the underlying emotions within the content. The system utilizes advanced Natural Language Processing (NLP) techniques, including transformer-based models, for text simplification and summarization, ensuring that learning materials are presented in a clear and accessible manner. Additionally, an emotion classification module analyzes the text to identify its emotional tone (e.g., happiness, sadness, excitement), which is then used to generate voice output modulated to reflect the detected emotions. By integrating these capabilities, the system creates emotionally expressive, personalized audio narratives that enhance comprehension, engagement, and retention for children with cognitive disabilities. The expected outcome of this research is the development of an innovative, adaptive educational platform that bridges the gap between accessibility and emotional connection, fostering an inclusive learning environment. The findings will contribute to more empathetic and effective assistive learning technologies, supporting diverse educational needs in both academic and therapeutic contexts.

Keywords: Text Simplification, Text Summarization, Emotion Detection, Natural Language Processing (NLP), Transformer Models, Emotion-Aware Speech Synthesis, Assistive Learning Technologies, Cognitive Disabilities, Accessible Education, Personalized Storytelling, Voice Modulation.

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LIST OF ABBREVIATIONS

Keyword	Meaning
NLP	Natural Language Processing
TTS	Text to Speech
BERT	Bidirectional Encoder Representations from Transformers
OCR	Optical Character Recognition – technology to extract text from images.

1. INTRODUCTION

Inclusive education plays a vital role in ensuring that children with cognitive disabilities have equal access to learning opportunities. For many of these children, traditional reading materials and teaching methods can be challenging due to complex vocabulary, lengthy content, and a lack of emotional engagement. As a result, they may face difficulties in understanding context, retaining information, and maintaining interest in learning activities. These challenges can impact not only academic performance but also emotional well-being and self-confidence.

Existing educational tools often rely on generalized content delivery methods, which fail to address the unique comprehension needs of children with cognitive impairments. While some assistive technologies exist, they tend to focus on either text simplification or audio-visual support, lacking an integrated approach that combines comprehension enhancement with emotional engagement. This limitation creates a gap in providing fully adaptive learning experiences tailored to the learner's needs.

The current gap lies in the absence of an intelligent, multi-modal learning system that can simplify complex text, generate concise summaries, and deliver content in an emotionally expressive manner. While there are advances in Natural Language Processing (NLP) and speech synthesis, most systems do not adapt storytelling based on the learner's cognitive level or emotional context. This research aims to bridge this gap by developing a solution that leverages NLP techniques for text simplification and summarization, integrates emotion detection models for adaptive storytelling, and utilizes text-to-speech (TTS) technology for engaging auditory delivery.

The primary objective of this research is to design and implement an NLP-powered educational assistant capable of transforming reading material into simplified, emotionally rich narratives. The system will analyze text complexity, reduce it to an appropriate reading level, and produce summaries that preserve key ideas while minimizing cognitive load. Additionally, the TTS module will incorporate emotional prosody to make the listening experience more engaging, fostering better comprehension and emotional connection to the content.

By delivering content that is both cognitively accessible and emotionally engaging, this research has the potential to significantly improve literacy skills, comprehension, and overall learning outcomes for children with cognitive disabilities. The findings from this study will contribute to the development of more inclusive educational technologies, paving the way for adaptive learning systems that can be applied in classrooms, therapy sessions, and home learning environments.

Given the growing recognition of personalized education, this research represents an important step toward bridging accessibility gaps and creating learning tools that truly adapt to the needs of every child. The proposed system aims to enhance educational inclusion and provide a scalable solution for learners with diverse abilities across various educational settings.

1.1 Background & Literature survey

Children with cognitive impairments face distinct challenges in language processing, comprehension, and emotional engagement, which often hinder their ability to fully participate in traditional storytelling activities [1]. Such impairments may involve difficulties with memory, attention span, problem-solving, and recognizing emotions, making it harder for learners to follow complex narratives [9]. Storytelling, however, remains a vital educational tool, as it not only fosters vocabulary growth and comprehension skills but also nurtures imagination, empathy, and emotional intelligence [10].

Despite its benefits, conventional storytelling methods—such as reading directly from books or using unmodified digital e-books—are typically designed for neurotypical learners and lack adaptive features to cater to diverse cognitive needs [5]. As a result, children with cognitive impairments may lose interest quickly, fail to grasp the storyline, or struggle to connect emotionally to characters and events [6].

Advances in Natural Language Processing (NLP), speech synthesis, and affective computing have opened new opportunities for creating intelligent, adaptive learning environments [3]. By integrating text simplification, summarization, and emotion detection, it is possible to design a dynamic storytelling system that personalizes the learning experience in real time [9].

In the proposed system, the first core module—**text simplification**—modifies complex vocabulary and sentence structures into more accessible forms without losing meaning or emotional depth [1]. The second module—**text summarization**—condenses long passages into shorter versions, improving comprehension for learners with limited attention spans [9].

The third and most innovative module is **emotion detection**, implemented using facial expression recognition through Convolutional Neural Networks (CNNs) [7]. By capturing and processing real-time video of the learner’s face, the system can assess emotional states such as happiness, confusion, or disengagement [7]. This feedback allows the storytelling system to adapt its delivery dynamically—adjusting tone, pace, pitch, and expressiveness of the narration. For example, if the child appears disengaged, the system can switch to a more energetic and visually rich storytelling mode [8].

To ensure a **multi-sensory learning experience**, the system will integrate speech synthesis with emotional modulation, enabling the narrator’s voice to convey the intended emotions of the story’s characters [3]. Visual cues, such as animated expressions or highlighted text, will further reinforce meaning and emotional impact [4]. This combination of auditory, visual, and affective feedback aims to improve comprehension, sustain engagement, and enhance emotional literacy among children with cognitive impairments [5].

Technically, the framework will require real-time facial expression analysis, NLP-based content adaptation, and speech synthesis APIs [7]. Emotion recognition may leverage datasets such as AffectNet or CK+, trained on diverse facial expressions [7], while text simplification models may use corpora like Newsela or Simple Wikipedia [1]. NLP modules will likely employ transformer-based architectures (e.g., BERT, T5) for semantic understanding and summarization [9], while CNN-based architectures will process visual input for emotion recognition [7].

The adaptive learning loop will operate in three stages:

1. **Input analysis** — Capturing story text and learner’s facial expressions.
2. **Content adaptation** — Applying text simplification, summarization, and emotional tuning.
3. **Dynamic delivery** — Narrating the story with emotion-adjusted speech synthesis and visual support.

By incorporating affective computing into educational software, this approach not only personalizes content but also makes the educational process emotionally intelligent [3]. Such a system can be deployed in classrooms, therapy sessions, or at home, supporting inclusive education and bridging the gap between learners with and without cognitive impairments [10].

In conclusion, the proposed **Emotionally Expressive Story Reading System** has the potential to revolutionize how children with cognitive impairments engage with stories. Through the combination of NLP-driven text adaptation, CNN-powered emotion recognition, and expressive speech synthesis, it promotes accessibility, engagement, and emotional development, contributing to equitable learning environments [1], [3], [7].

1.2 Research Gap

Identifying research gaps is a critical element of academic inquiry, as it highlights areas that remain underexplored and provides justification for the proposed work. In the context of the proposed **NLP-based module**—which integrates text simplification, summarization, and emotion extraction for children with cognitive impairments—a review of the literature reveals significant opportunities for innovation. Specifically, the integration of adaptive text simplification with real-time summarization, coupled with fine-grained emotion detection, remains largely unexplored in accessibility-focused educational technologies.

The first study in the literature [8] investigates advanced text simplification techniques using Transformer-based architecture such as **BERT** and **T5**. While the results demonstrate substantial improvements in reducing syntactic and lexical complexity without compromising semantic integrity, the approach does not address **dynamic adaptation** to the cognitive levels of diverse user groups. No personalization strategies were proposed for learners with **visual, hearing, or cognitive disabilities**, leaving a gap in **real-time complexity adjustment** based on individual comprehension profiles.

The second work [7] focuses on **abstractive summarization** using models such as **PEGASUS** and **Longformer**. The study achieves improved contextual coherence and factual accuracy, yet it does not incorporate **accessibility considerations** or preserve **emotional tone**, both of which are crucial in educational storytelling. This omission highlights a research gap in **integrating summarization techniques with emotion-sensitive delivery**, particularly for children whose engagement and comprehension are strongly tied to emotional context.

The third study [9] explores **emotion detection** in text using hybrid approaches that combine lexicon-based methods with deep learning classifiers, including **CNNs** and **BiLSTMs**. While achieving high accuracy in classifying discrete emotions such as joy, sadness, anger, and surprise, the study does not address **multi-label emotion recognition** in multi-sentence contexts, nor the challenge of aligning emotional cues with adaptive simplification and summarization tasks.

From this analysis, it is evident that while **text simplification**, **summarization**, and **emotion detection** have each been studied extensively, they have largely been addressed **in isolation** [6]–[9]. Existing works typically focus on **adult-level comprehension** or **standard readability metrics**, neglecting the needs of **children with cognitive impairments**, who require both **linguistic accessibility** and

emotional engagement for effective learning [1], [3].

The proposed research seeks to bridge these gaps by developing an **integrated NLP framework** that:

1. **Combines adaptive text simplification** with context-aware summarization.
2. **Incorporates fine-grained emotion detection** for personalized emotional feedback.
3. **Uses user profiling** to dynamically tailor content complexity and emotional tone in real time.

This unified system will leverage **transformer-based semantic models** for linguistic adaptation and **CNN-based facial expression recognition** for emotional analysis [6], [9]. The intended outcome is an **inclusive educational storytelling platform** that not only improves **comprehension** but also fosters **emotional connection**, enabling equitable learning opportunities for children with cognitive impairments [1], [3].

contributing to improved learning experiences for diverse user groups.

Research Gap Feature	[6]	[7]	[8]	[9]	Proposed Research Solution
Adaptive text simplification tailored for children with cognitive impairments	✗	✗	✓	✗	✓
Real-time summarization aligned with attention span and accessibility	✗	✓	✗	✗	✓
Emotion-sensitive summarization that preserves emotional tone	✗	✗	✗	✗	✓
Fine-grained emotion detection across multi-sentence contexts	✗	✗	✗	✓	✓

Table 1 -Research gap for component

1.3 Research Problem

Cognitive disabilities encompass a broad range of conditions that affect a person's ability to learn, process information, remember, communicate, and adapt to new situations. Children with cognitive impairments often face significant barriers in accessing educational content that matches their learning pace, comprehension ability, and communication needs. Traditional educational systems, whether in physical classrooms or digital platforms, are often designed with a one-size-fits-all approach, which fails to address the unique learning challenges these children encounter. This lack of adaptive and inclusive design leads to reduced engagement, slower knowledge acquisition, and, in many cases, frustration and withdrawal from the learning process.

One of the major issues in the education of children with cognitive disabilities is the complexity of the language used in learning materials. Standard curriculum content often contains long sentences, abstract concepts, and domain-specific vocabulary that are difficult for these children to understand. While special education teachers can manually simplify text, this is time-consuming and not feasible for all content or in real-time classroom situations. Furthermore, most e-learning platforms lack automated text simplification or summarization capabilities that are tailored to the comprehension level of children with cognitive impairments.

In addition to linguistic complexity, another barrier is the absence of emotionally expressive and multimodal feedback. Children with cognitive disabilities often require multisensory engagement—such as emotional cues in voice, relevant imagery, and interactive elements—to sustain attention and aid comprehension. Current digital reading tools frequently offer text-to-speech (TTS) capabilities, but these are often monotone and lack emotional modulation, which reduces the listener's ability to interpret tone, intent, and narrative mood. Research shows that emotional prosody in speech significantly enhances comprehension and memory retention, especially in young learners and those with learning challenges.

Moreover, there is a gap in context-aware semantic tagging that would enable software to provide real-time support, such as highlighting key concepts, defining difficult words, and linking them to visual or sign language representations. Without such features, children with cognitive disabilities struggle to connect abstract ideas with concrete understanding. While assistive technologies like screen readers and captioning exist, they are not optimized for cognitive accessibility and often focus solely on sensory disabilities, such as visual or hearing impairments.

Another challenge is the lack of personalized adaptability in current educational tools. Cognitive disabilities vary in type and severity—what works for one learner may not work for another. An effective system must

be able to adjust reading complexity, speech rate, emotional tone, and the density of information in real time, based on the learner's needs and responses. However, most current solutions are static and cannot dynamically adapt to the learner's progress or emotional state during the reading process.

The problem is further exacerbated by the lack of integration between different assistive features. For example, a tool may provide text simplification, but without synchronized expressive TTS or visual aids, the benefits are limited. Similarly, sign language representation may exist for some words, but without semantic context detection, its use is sporadic and inconsistent. The absence of a holistic, integrated system means children must switch between multiple platforms or tools, leading to cognitive overload and reduced learning efficiency.

Finally, there is a communication gap between educational software and caregivers or teachers. Without real-time feedback on a child's reading performance, comprehension level, or engagement patterns, it becomes difficult to provide targeted intervention. Most current platforms lack analytics and reporting features that could help educators identify which parts of the text caused confusion or disengagement.

Given these gaps, there is a pressing need for an inclusive, intelligent educational software system that integrates natural language processing (NLP) for text simplification and summarization, emotion detection for adaptive TTS modulation, context-aware semantic tagging for visual and sign language support, and real-time adaptability for personalized learning experiences. Such a system should also incorporate analytics for teachers and caregivers, enabling targeted educational interventions. Addressing these problems would not only enhance reading comprehension for children with cognitive disabilities but also contribute to their emotional engagement, self-confidence, and long-term academic success.

2. RESEARCH OBJECTIVES

2.1 Main Objectives

The main objective of this study is to develop a comprehensive cognitive assistance system that integrates advanced Natural Language Processing (NLP) techniques with real-time emotion recognition and adaptive content delivery to support children with disabilities, including visual, hearing, and cognitive impairments. The system will utilize deep learning models for text simplification, summarization, and semantic tagging, along with multimodal technologies such as text-to-speech (TTS) and sign language generation, to enhance accessibility and comprehension. The overall aim is to create a scalable solution that not only detects cognitive and emotional challenges during reading activities but also provides personalized, context-aware support in real time, thereby improving learning outcomes, engagement, and inclusivity in educational environments.

2.2 Specific Objectives

To achieve the main objective, the study will focus on the following specific objectives:

1. **Accurate Emotion Detection from Text for Children with Cognitive Disabilities:**

Develop an advanced NLP-based system capable of accurately identifying emotional states from educational and story-based text materials. This objective is Specific as it targets the emotional understanding aspect for cognitive learning, Measurable by evaluating precision, recall, and F1-scores of emotion detection models, Achievable through the use of state-of-the-art algorithms and annotated datasets, Realistic given the current advancements in affective computing and NLP, and Time-bound by defining a structured development and testing timeline.

2. **Automated Text Simplification for Enhanced Comprehension:**

Implement a text simplification module to reduce linguistic complexity while retaining core meaning, ensuring content is accessible to children with varying levels of cognitive ability. This objective is Specific in targeting language simplification, Measurable by readability indices (e.g., Flesch–Kincaid score) and comprehension test results, Achievable through machine learning-based simplification techniques, Realistic given the availability of pre-trained models, and Time-bound with defined milestones for integration into the system.

3. Summarization of Educational and Story Texts:

Design an automated summarization feature to condense lengthy texts into short, meaningful summaries suitable for learners with limited attention spans or memory retention challenges. This objective is Specific to the summarization process, Measurable by summary quality metrics such as ROUGE and BLEU scores, Achievable with transformer-based summarization models, Realistic within the scope of the research, and Time-bound with specific evaluation and refinement phases.

4. Context-Aware Semantic Tagging for Personalized Learning:

Integrate semantic tagging mechanisms that label simplified or summarized texts with contextually relevant keywords, emotions, or difficulty levels to support personalized learning pathways. This objective is Specific in targeting adaptive learning, Measurable by accuracy in tagging and user satisfaction scores, Achievable through NLP pipelines combining named entity recognition and emotion classification, Realistic within available computational resources, and Time-bound with a planned implementation period.

5. Evaluation of Impact on Learning Outcomes:

Assess the effectiveness of the text simplification, summarization, and emotion detection modules in improving comprehension, engagement, and emotional recognition skills of children with cognitive disabilities and their feedbacks can construct and keep the pace of education. This objective is Specific in evaluating impact, Measurable through pre- and post- intervention assessments, Achievable by collaborating with special education institutions for pilot testing, Realistic in terms of data collection and analysis, and Time-bound with scheduled evaluation checkpoints.

By fulfilling these objectives, the study seeks to develop an inclusive educational NLP module that bridges the gap between complex educational content and the learning needs of children with cognitive disabilities, and receiving the feedbacks so that this system can fostering both comprehension and emotional development.

3. METHODOLOGY

3.1 Key Pillars in the Specialized Area

The methodology for the proposed inclusive educational software system includes several integrated components, each designed to enhance learning outcomes for children with cognitive disabilities through accessible, adaptive, and engaging content delivery. The key pillars of this research lie in the application of Natural Language Processing (NLP) techniques for text simplification, summarization, emotion detection, and context-aware semantic tagging. These techniques are well-suited for improving comprehension and engagement across diverse learning needs. The methodology will emphasize developing and applying these technologies to achieve the objectives of content accessibility, real-time adaptability, and multimodal delivery, ensuring an inclusive learning experience that supports varied cognitive abilities effectively.

3.2 Proposed Approach and Task Breakdown

1. Violation Detection

Task Overview:

Content adaptation forms the foundation of the inclusive educational software system, focusing on tailoring learning materials to meet the needs of children with cognitive disabilities. This task involves a comprehensive approach combining advanced Natural Language Processing (NLP) techniques and emotion detection to ensure content is accessible, engaging, and appropriate for each learner. The system processes text inputs and learning materials to simplify language, summarize key points, and incorporate emotional cues that enhance understanding. This step ensures that content is adapted in real time, enabling the system to deliver inclusive, personalized learning experiences that accommodate diverse cognitive needs.

Sub-Tasks:

- **Text Simplification and Summarization:** This involves processing educational content to reduce linguistic complexity while preserving meaning. NLP techniques, such as syntactic parsing, lexical simplification, and abstractive summarization, are applied to make content easier to read and comprehend for learners with cognitive impairments.
- **Emotion Detection and Context-Aware Semantic Tagging:** Emotion detection uses machine learning models to analyze the tone, sentiment, and context of text or speech to identify the emotional state of the learner. Context-aware semantic tagging further enriches content by highlighting important concepts and adapting instructional strategies, ensuring the learning experience is responsive, supportive, and personalized.

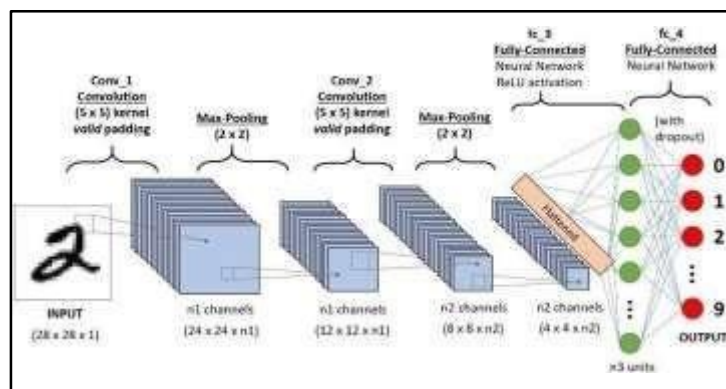


Figure 1 Common Convolutional Neural Network

- **Assistive Technology Integration:**

Assistive technologies, such as screen readers, text-to-speech (TTS) engines, and adaptive user interfaces, will provide additional support for learners with cognitive disabilities. These tools can enhance the effectiveness of the NLP-based content adaptation by offering multiple modes of content delivery, such as auditory, visual, and interactive formats. This integration is particularly valuable in situations where reading comprehension may be limited, attention may fluctuate, or learners require multimodal reinforcement. By combining these assistive tools with simplified and emotionally adaptive content, the system ensures accessibility, engagement, and inclusivity for a wide range of cognitive learning needs.

2. Real-Time Monitoring

Task Overview:

Real-time monitoring is essential for ensuring the system continuously analyzes text input and emotional cues, enabling accurate and timely adaptations to suit children with cognitive disabilities. This task involves the ongoing capture and processing of data from text inputs, context detection modules, and emotion recognition algorithms. The monitoring system evaluates this data in real-time, comparing it against predefined accessibility and comprehension parameters. Any deviations—such as overly complex text, high cognitive load, or negative emotional indicators—are immediately flagged for adjustment.

The real-time nature of this process ensures the system remains responsive and adaptive, providing immediate support to the learner's needs.

Sub-Tasks:

- Continuous Data Capture: The system will continuously collect text data, contextual cues, and user interaction feedback. This involves integrating modules for natural language input, emotion detection, and real-time user engagement tracking.
- Data Processing: Captured data will be processed using NLP algorithms, text simplification, and sentiment/emotion analysis to detect comprehension barriers or emotional distress.
- Monitoring Software: A central processing module will manage real-time analysis across multiple components, enabling instant decision-making to adjust text complexity, tone, or delivery method.

3. User Notification**Task Overview:**

This task involves developing a user-facing interface that delivers real-time adaptive feedback to the learner and support notifications to caregivers or educators. Upon detecting comprehension issues or emotional challenges, the system sends targeted prompts, simplified explanations, or alternative delivery formats such as audio narration or visual aids. Caregivers receive relevant insights to help guide the learner effectively.

Sub-Tasks:

- Notification System Development: A responsive interface (web or mobile) will be created to deliver both learner and caregiver notifications. It will integrate with the NLP and emotion detection modules to trigger alerts instantly.
- Alert Mechanism: Learners will receive adaptive on-screen prompts such as “Would you like an easier version of this text?” while caregivers receive progress and assistance alerts, ensuring immediate support and intervention when needed.

4. Feedbacks Response Assistance

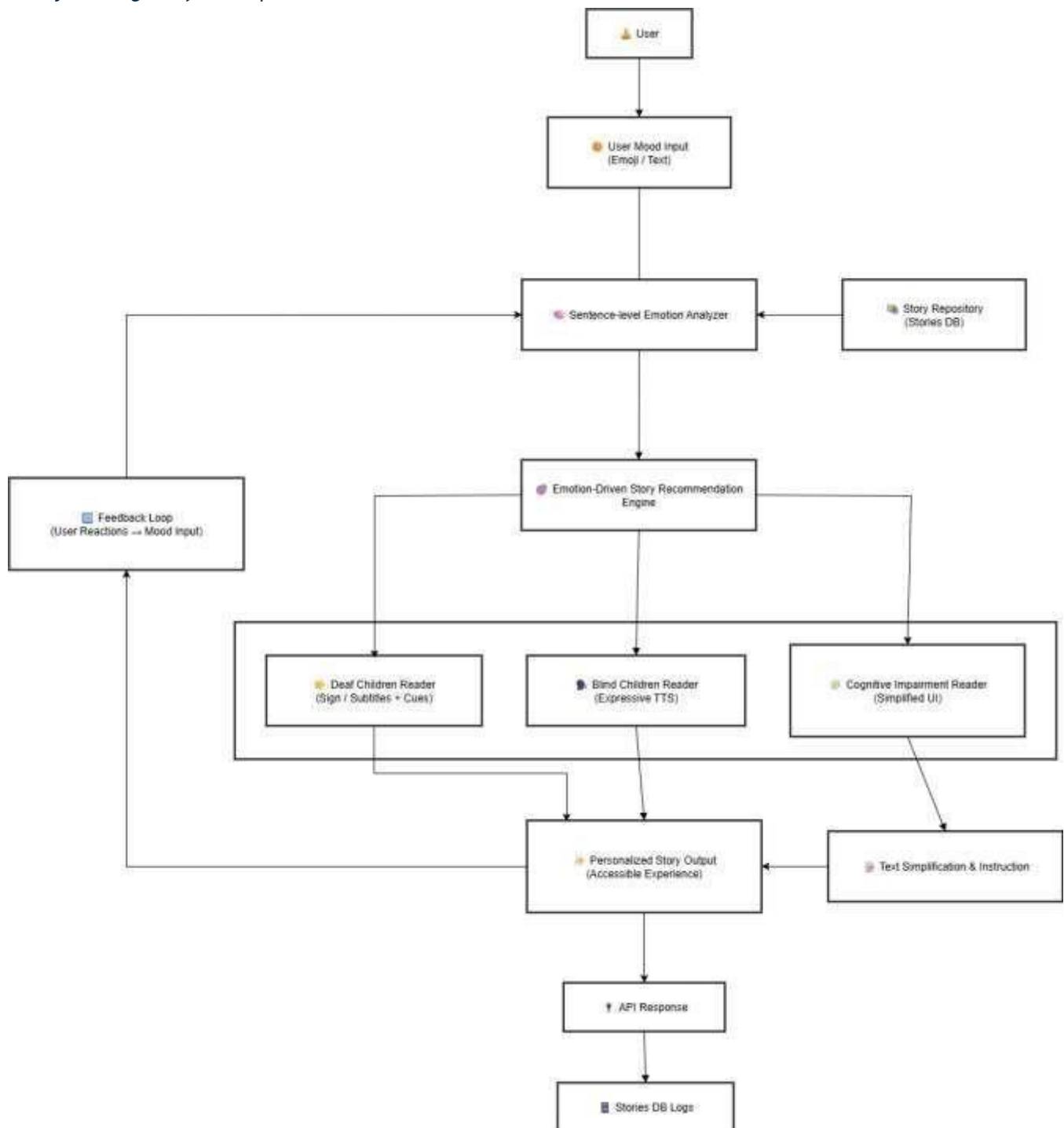
Task Overview:

This task focuses on receiving the personalized feedback when comprehension challenges persist despite real-time adjustments. The system categorizes feedbacks and emotional states, then assigns an appropriate adaptive response—ranging from simplified rewording to switching content into visual or audio formats for further improvement. This process is automated and designed to ensure learners consistently receive accessible and leasable, emotionally supportive content.

Sub-Tasks:

- **Difficulty & Emotion Categorization:** Interaction and emotion data are categorized to determine the severity of comprehension barriers or emotional discomfort.
- **Automated Feedback Delivery:** When a challenge persists, the system automatically delivers feedback receivables for additional explanations tailored to the learner’s cognitive needs.
- **Assistance Management System:** An integrated log will track support provided, monitor its effectiveness, and provide the feedbacks received and it’ll improve further personalization and strategies.

Figure 2 - System Diagram of the component



3.4 Proposed Project Plan

Data collection will be crucial in developing and validating the NLP module of the Inclusive Educational Software System, particularly in training the text simplification, summarization, emotion detection, and semantic tagging models. The project will require a comprehensive dataset, gathered from various sources, to cover all aspects of cognitive disability–friendly content generation, including linguistic variations, emotional tones, and comprehension levels. This data will be collected from educational resources, accessibility-focused datasets, and possibly supplemented with insights from educators and therapists.

1. Text Data for Model Training

Data Needed: A diverse corpus of educational text materials in multiple complexity levels, covering different subjects, sentence structures, and emotional tones. The dataset should include examples tailored for children with varying levels of cognitive abilities.

Collection Method: Educational content will be sourced from open-access learning platforms, government- provided special education resources, and curated books designed for learners with disabilities. Text will be annotated for complexity level, emotional tone, and key concepts, enabling accurate training of the text simplification, summarization, and semantic tagging models.

2. Real-Time Interaction Data

Data Needed: Live input from users during system use, including the original educational text entered, the simplified output, and feedback on clarity and comprehension.

Collection Method: The software will log user interactions in real time, capturing both the original and processed content. This feedback loop will allow the system to validate the accuracy of its NLP outputs and adapt to individual learning needs dynamically.

3. Survey and Interview Data

Data Needed: Insights from educators, therapists, caregivers, and students regarding challenges in understanding educational content, preferred learning styles, and emotional engagement factors.

Collection Method: Surveys and structured interviews will be conducted with stakeholders in special education. This qualitative data will help refine model outputs to ensure they are accessible, emotionally engaging, and cognitively suitable for the target audience.

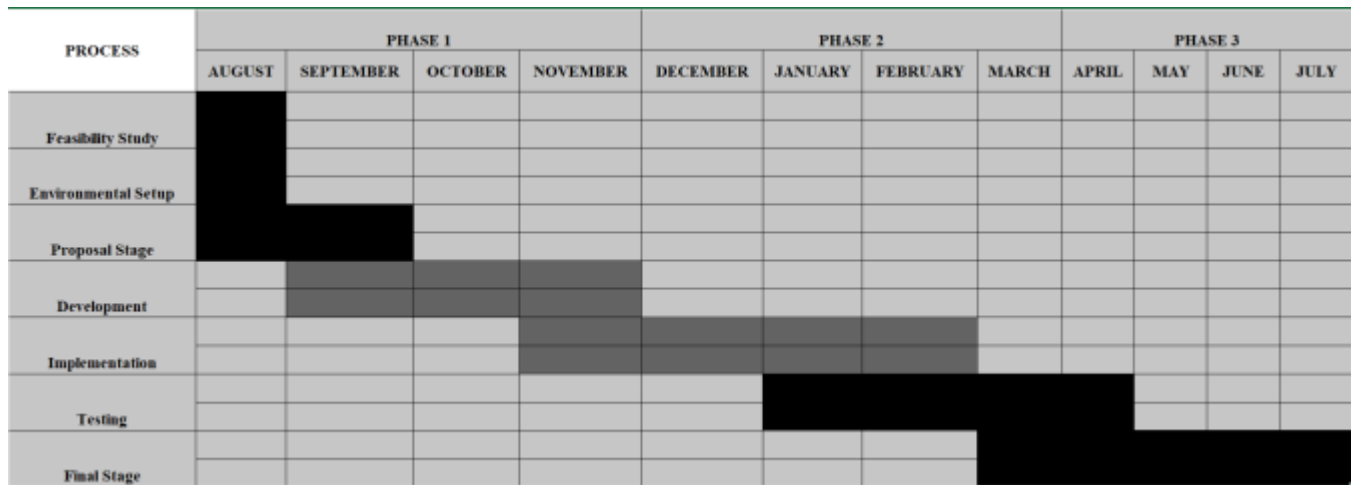


Figure 3 - Gantt Chart



Figure 4- Work Breakdown Structure

3.5 Anticipated Conclusion

3.5.1 Results

The inclusive educational software developed through this study is expected to significantly enhance the learning experience of children with cognitive disabilities by integrating advanced natural language processing (NLP) techniques with real-time accessibility features. By leveraging modules such as text simplification, summarization, emotion detection, and context-aware semantic tagging, the system will deliver content that is both understandable and engaging for learners. The continuous real-time adaptation will ensure that learning materials are tailored to each child's comprehension level and emotional state as the session progresses. Automated assistive outputs will follow, including speech synthesis for auditory learning and sign language animations for visual comprehension, ensuring consistent accessibility for diverse learners. Collectively, these components will lead to improved understanding, increased engagement, and more inclusive educational experiences for children with cognitive disabilities.

3.5.2 Application

The system is applicable across various educational and assistive learning environments where effective support for children with cognitive disabilities is essential. It is ideally suited for inclusive classrooms, special education centers, and home-based learning setups, where adapting content to individual needs can significantly improve comprehension and engagement. Within the context of modern inclusive education initiatives, this system aligns with the goals of integrating advanced technologies to enhance accessibility and personalized learning. Institutions such as mainstream schools with inclusive programs, therapy centers, and community learning hubs can benefit from the system by streamlining lesson delivery, reducing the cognitive load on learners, and enhancing the overall educational experience by making content more understandable, engaging, and tailored to each child's needs.

3.5.3 Real World Use

In practical terms, the system will enhance inclusive education by improving the delivery of learning materials and adapting them to meet the needs of children with cognitive disabilities. In mainstream classrooms and special education centers, it will support large groups of learners by providing personalized content that accommodates varying comprehension levels, minimizing the risk of learners being left behind. At therapy centers and home-based learning environments, where individualized attention is critical, this system will offer a dependable solution for tailoring lessons and activities to each child's abilities. Additionally, as part of broader inclusive education initiatives, the system can be integrated into national and institutional frameworks, contributing to the development of intelligent learning infrastructure that supports equitable and sustainable educational growth.

3.6 Commercialization

The commercialization strategy for the “Inclusive Educational Software System” is designed to target key markets such as special education institutions, mainstream schools with inclusive programs, therapy centers, and home-based learning environments. These sectors represent a significant demand for innovative educational tools that can enhance accessibility, improve learning outcomes, and provide personalized experiences for children with cognitive disabilities. For educational institutions, the system presents considerable economic and academic advantages by reducing the reliance on extensive manual lesson customization, which in turn lowers teacher workload and resource costs. The platform’s sophisticated content adaptation and progress tracking mechanisms ensure targeted learning, reducing inefficiencies and minimizing the risk of educational neglect. This design also allows for future upgrades and integration with emerging assistive technologies, ensuring long-term relevance and flexibility.

Economically, the Inclusive Educational Software System offers substantial value by optimizing teaching resources, lowering operational expenses, and improving measurable learning outcomes. Its adaptive and modular design ensures flexibility and scalability, making it suitable for a wide range of environments, from small classrooms to nationwide education networks.

By focusing on these core markets and emphasizing the dual benefits for both learners and educators, the Inclusive Educational Software System is well-positioned to become a leading solution in the field of advanced assistive educational technology.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Functional Requirements outline the specific features and functionalities that the inclusive educational software must provide:

- **Emotion-Aware Text Simplification:** The system must automatically simplify educational text based on the child's cognitive level while preserving the intended emotional tone and meaning. It should use Natural Language Processing (NLP) techniques to ensure age-appropriate vocabulary and sentence structure.
- **Context-Aware Summarization:** The system should generate concise, child-friendly summaries of long passages, ensuring that essential ideas are retained and presented in a clear, understandable manner.
- **Emotion Detection and Feedback:** The system should analyze the child's emotional responses (captured via text, audio, or facial expressions) to adjust content difficulty, tone, and delivery in real time, ensuring an emotionally supportive reading experience.
- **Multimodal Output Delivery:** The system must support multiple output formats such as text-to-speech (with expressive voice modulation), sign language animations, and visual cues to cater to various disabilities.

4.2. User Requirements

User requirements describe the needs and expectations of the system's end-users:

- **Ease of Use:** The interface should be simple, intuitive, and engaging, allowing children with cognitive disabilities to navigate without unnecessary complexity.
- **Personalization:** The system should adapt reading content, pacing, and presentation style to each child's unique learning profile and emotional state.
- **Accessibility:** Users must be able to interact with the system through various assistive technologies, including screen readers, alternative input devices, and sign language support.
- **Engagement & Motivation:** The system should use gamification elements, encourage progress and maintaining the learner's interest during reading activities.

4.3. System Requirements

System Requirements outline the technical specifications and components needed to implement the inclusive educational software:

- **Hardware:** Devices such as tablets, computers, or smartboards with microphones, speakers, and cameras for capturing user input and delivering multimodal content.
- **Software:** NLP-based text simplification and summarization modules, emotion detection algorithms, TTS engines with expressive voice synthesis, and sign language rendering modules and feedbacks for the system improvements.
- **Database:** A secure database to store user profiles, reading history, emotional response patterns, feedbacks and progress tracking data.
- **Network:** Reliable internet connectivity for cloud-based processing, content updates, and integration with third-party assistive tools.

4.4. Non-Functional Requirements

Non-Functional Requirements specify the quality attributes and constraints of the system:

- **Accuracy:** The system must ensure high accuracy in simplifying text, detecting emotions, and delivering appropriate multimodal responses, minimizing misinterpretations.
- **Scalability:** The software should be scalable to support multiple users simultaneously, both in small classroom settings and large institutional deployments.
- **Reliability:** The system should maintain stable performance with minimal downtime, even when processing real-time multimodal inputs.
- **Performance:** Content processing, personalization, and response adaptation should occur in near real-time to maintain the flow of the learning session.

BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)
Web Hosting	9,600
Development Tools and Techniques	22,400
Cloud Hosting and Data Storage	12,000
Paper publishing cost	50,000
Total	94, 000

Table 2 - Budget justification

REFERENCES

- [1] M. Štajner and S. Ponzetto, “Automatic Text Simplification for People with Cognitive Disabilities,” *Frontiers in Artificial Intelligence*, vol. 4, pp. 1–14, May 2021. doi: 10.3389/frai.2021.624189.
- [2] R. Baeza-Yates, P. Boldi, and C. Castillo, “Characterizing accessibility issues in web search engines,” *Information Processing & Management*, vol. 56, no. 3, pp. 707–719, May 2019. doi: 10.1016/j.ipm.2018.12.003.
- [3] R. Martínez, M. Cuayáhuil, and R. Higashinaka, “Enhancing Text-to-Speech Systems with Emotional Prosody for Improved User Engagement,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, Brighton, UK, May 2019, pp. 7085–7089. doi: 10.1109/ICASSP.2019.8683186.
- [4] M. Camgoz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, “Neural Sign Language Translation,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, Jun. 2018, pp. 7784–7793. doi: 10.1109/CVPR.2018.00812.
- [5] A. Bharath and P. Morris, “Assistive Technologies for Visually Impaired Learners: A Review,” *British Journal of Visual Impairment*, vol. 39, no. 1, pp. 55–72, Jan. 2021. doi: 10.1177/0264619620941831.
- [6] P. V. V. Kishore, R. R. Kumar, D. A. Kumar, E. Kiran, and K. Kishore, “Indian Sign Language Recognition using Graph Matching on 3D Motion Captured Data,” *Multimedia Tools and Applications*, vol. 76, no. 3, pp. 4347–4375, Feb. 2017. doi: 10.1007/s11042-016-3790-5.
- [7] N. Hossain, M. A. Uddin, and S. Alam, “A Review of Emotion Recognition in Text for Sentiment Analysis,” *IEEE Access*, vol. 8, pp. 50689–50701, Mar. 2020. doi: 10.1109/ACCESS.2020.2975713.
- [8] A. Williams, R. K. Moore, and S. King, “Speech Synthesis for the Visually Impaired: Enhancing Storytelling with Expressive Prosody,” in *Proc. Interspeech 2020*, Shanghai, China, Oct. 2020, pp. 2102–2106. doi: 10.21437/Interspeech.2020-1521.
- [9] M. S. Syed, H. L. Seng, and A. H. Abdullah, “A Review on Text Simplification for Cognitive Disability,” *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 6, pp. 2986–3002, Jun. 2022. doi: 10.1016/j.jksuci.2019.06.005.
- [10] R. J. Williams, “Multi-modal Learning Interfaces for Children with Disabilities: A User-Centered Design Approach,” *ACM Transactions on Accessible Computing*, vol. 13, no. 4, pp. 1–28, Dec. 2020. doi: 10.1145/3431346.